

Subject: Engineering Chemistry

PART-A

1. What is degree of hardness?
2. What is caustic embrittlement?
3. Define boiler troubles?
4. Define scale & sludge formation & their consequences.
5. What is octane & cetane numbers?
6. What do you mean by 'Disinfection'?
7. Define viscosity & viscosity index?
8. What are constituents of Portland cement?
9. Why gypsum is added to cement?
10. What is annealing of glass and its importance?
12. Write Dulong's formula.
13. State Markownikov's rule with example.
14. Write the structure and IUPAC name of Paracetamol and Aspirin.
15. What is synthetic petrol?
16. Define breakpoint chlorination.
17. What is pilling-Bedworth rules?
18. What is sweetening of petrol?
19. Define glassy state of matter?
20. What are electrophile & nucleophile?

21. Why do we express hardness in terms of calcium carbonate equivalent?
22. What is carbonization of coal?
23. What is hard glass.
24. Define steam emulsion number.
25. What are consequences of corrosion.
26. Why should an ideal fuel have moderate ignition temperature?
27. Why impure metal corrode faster than pure metal under ideal conditions?
28. Define cloud & pour point.
29. What is stress corrosion?
30. Define knocking.

PART-B

1. Write the method for the preparation of synthetic petrol.
- X (2) Explain the method for the determination of calorific value of ~~solid~~ gaseous fuel.
3. Calculate the amount of lime and soda required to soften the 50,000 L of water sample containing the following impurities
 $\text{CaSO}_4 = 0.40 \text{ g/L}$; $\text{MgCO}_3 = 0.44 \text{ g/L}$; $\text{CaCO}_3 = 2.0 \text{ g/L}$
 $\text{MgSO}_4 = 0.80 \text{ g/L}$; $\text{MgCl}_2 = 0.78 \text{ g/L}$; $\text{SiO}_2 = 2.40 \text{ g/L}$
 & $\text{NaCl} = 2.50 \text{ g/L}$,
 if purity of lime is 90% and that of soda is 95%.

PART-B

③

- (4) Describe the proximate analysis of coal and its significance.
- ⑤ Describe zeolite method for the softening of water.
- ⑥ Describe synthesis properties and uses of Aspirin.
- ⑦ Describe the mechanisms of lubrication.
- ⑧ Explain the mechanism of Wet corrosion
- ⑨ Write the mechanism of free radical reactions of alkanes.

PART-C

1. Describe the ion-exchange method for the purification (softening) of water.
- ② Write the process & chemical reaction for the setting & hardening of Portland cement.
3. Explain the mechanism of :-
 - (a) ~~Free radical~~ Elimination in alkyl halide
 - (b) Nucleophilic addition in carbonyl compounds (aldehyde & ketones)
4. Write the methods for the protection of metal from corrosion.
5. Write short notes on:
 - ① Galvanic corrosion
 - ⑥ Concentration cell corrosion

(X) ⑥ Describe Otto-Hoffmann by-product coke oven method. ④

⑦ What is cracking? Explain fixed bed catalytic cracking.

Numericals :-

1. A sample of hard water has hardness 500 ppm. Calculate the hardness in °French & °Clark.
2. A sample of water contains 400 mg of CaCO_3 per litre. Calculate the hardness in terms of CaCO_3 equivalent.
3. A sample of water has following impurities
 $\text{Mg}(\text{HCO}_3)_2 = 14.6 \text{ mg/L}$; $\text{MgCl}_2 = 19 \text{ mg/L}$;
 $\text{MgSO}_4 = 36 \text{ mg/L}$; $\text{Mg}(\text{NO}_3)_2 = 29.6 \text{ mg/L}$
 $\text{NaCl} = 29 \text{ mg/L}$.
Calculate temporary, permanent & total hardness in ppm & °Clark.
4. A standard hard water contains 15 gm of CaCO_3 per litre, 20 ml. of this required 25 ml. of EDTA sol., 100 ml. of sample water required 18 ml. of EDTA sol. and 100 ml. of boil water of this sample water required 12 ml. of EDTA sol.. Calculate the temporary, permanent and total hardness of sample water.
5. A sample of coal containing $\text{C} = 75\%$;
 $\text{H}_2 = 8\%$; $\text{O}_2 = 7.5\%$; $\text{S} = 5.0\%$ & rest is ash.
Calculate the gross and net calorific value.

Numericals

6. A gaseous fuel have the following composition by volume :-

$H_2 = 20\%$, $CH_4 = 5\%$; $CO = 20\%$; $CO_2 = 5\%$.

& $N_2 = 50\%$.

If 50% excess air is supplied, calculate the volume of air required per m^3 of this gas. (fuel).

7. A sample of coal contain $C = 93\%$, $H = 6\%$ & remaining ash. The following data were obtained when the above coal was tested in bomb calorimeter :-

(i) Weight of coal = 0.92 gm.

(ii) Weight of water taken = 550 gm.

(iii) Water equivalent of bomb & calorimeter = 2200 gm.

(iv) ~~Raise~~ Rise in temp. = $2.42^\circ C$

(v) Fuse wire correction = 10.0 cal

(vi) Acid correction = 50.0 cal

Calculate the gross & net calorific value of the coal.

8. Describe lime-soda method for the softening of water.

9. What is phosphate conditioning? (B&T-A)